ECE 101L Lab4

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June 2026

1 Passive Low Pass Filter

1.1 Introduction

A low-pass filter is a circuit that, when a voltage is applied, will pass voltage only when the frequency of the input is low relative to a cut-off frequency. "Passive" indicates the exclusion of op amps and other active components.

The simplest low-pass filter is an input in series with a capacitor and a resistor, where the output is the voltage across the capacitor. Since the capacitors' impedance has a factor of the reciprocal of frequency $(\frac{1}{\omega})$, this is what causes it to pass only low frequencies.

We can model the output of the circuit at many frequencies by using something called the transfer function, $H(j\omega)$, which represents the circuit's gain as a function of frequency. We can also use the ratio of imaginary to real numbers to predict the output phase shift by taking its negative inverse tangent.

For our circuit, we should find a range of low frequencies where the output remains relatively level, then a range where the output starts to decay rapidly. We should also find that the phase shift will rise with frequency.

1.2 Experiment

The circuit consists of a $2k\Omega$ resistor and $10\mu F$ capacitor. Our input will be a sine wave of 1 volt peak to peak and variable frequency. The lab instructions called for a $1\mu F$ capacitor, but I accidentally took the wrong kind since it was put in the wrong place. I compensate for this mistake in my calculations. I made a similar mistake with the resistor, where the manual called for a $1k\Omega$ resistor. The circuit is pictured in Fig 1.

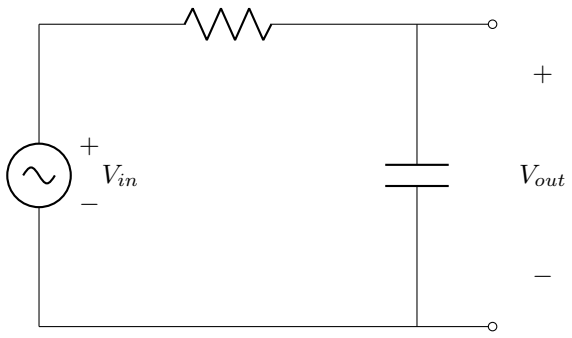


Figure 1: Low Pass Filter

The two main data points we are collecting are the peak-to-peak amplitude at each frequency and the time delay between the input and the capacitor. The time delay is essentially the time it takes for a signal to reach the capacitor and can be measured by observing the difference in time between landmark points of the input and output wave.

From there, we can also find the magnitude of gain, $|H(j\omega)|$, which is the output amplitude over the input amplitude. Since the input is approximately one volt peak to peak, this will equal the output amplitude. We can then find the gain magnitude in decibels by using the formula $20\ log(|H(j\omega)|)\ dB$

We can find the phase shift by using the formula $\phi^\circ = -360 \cdot f \cdot t_d$, where t_d is the time delay.

Table 1 displays all this information as measured from the circuit.

f	$V_{out}(V_{pp})$	$\lvert H(j\omega)\rvert$	$\lvert H(j\omega)\rvert\ \ (dB)$	$t_d\ \ (\mu s)$	$\phi\ \ (^\circ)$
$10Hz$	1	1	0	1000	-4
$50Hz$	0.975	0.975	-0.22	800	-14
$100Hz$	0.854	0.854	-1.37	660	-24
$160Hz$	0.733	0.733	-2.70	680	-39
$200Hz$	0.608	0.608	-4.32	570	-41
$400Hz$	0.422	0.422	-7.49	450	-65
$800Hz$	0.219	0.219	-13.19	260	-75
$1.6kHz$	0.107	0.107	-19.41	140	-81

Table 1: Low Pass Data

In order to create predictions for the phase and the magnitude, we can analyze the circuit and find its transfer function. We can start by using the voltage divider formula with impedance, then dividing both sides by V_{in}. This gives us a gain expression and will be equal to the transfer function. After that, we can multiply the numerator and denominator by $j\omega C$ to simplify the expression.

$$H(j\omega) = \frac{V_{out}}{V_{in}} = \frac{Z_C}{R + Z_C} = \frac{(\frac{1}{j\omega C})}{R + (\frac{1}{j\omega C})} = \frac{1}{Rj\omega C + 1}$$

To find a function for the phase shift from this, take the angle of the function with the function $\theta = arctan(\frac{Imaginary}{Real})$. Using the angle of quotients rule, $\angle\frac{A}{B} = \angle A - \angle B$.

$$\phi^\circ = arctan(\frac{0}{1}) - arctan(\frac{R\omega C}{1}) = -arctan(R\omega C)$$

Now knowing both of these functions, we can graph them and compare them to our found data. The graphs are pictured in figures 2 and 3.

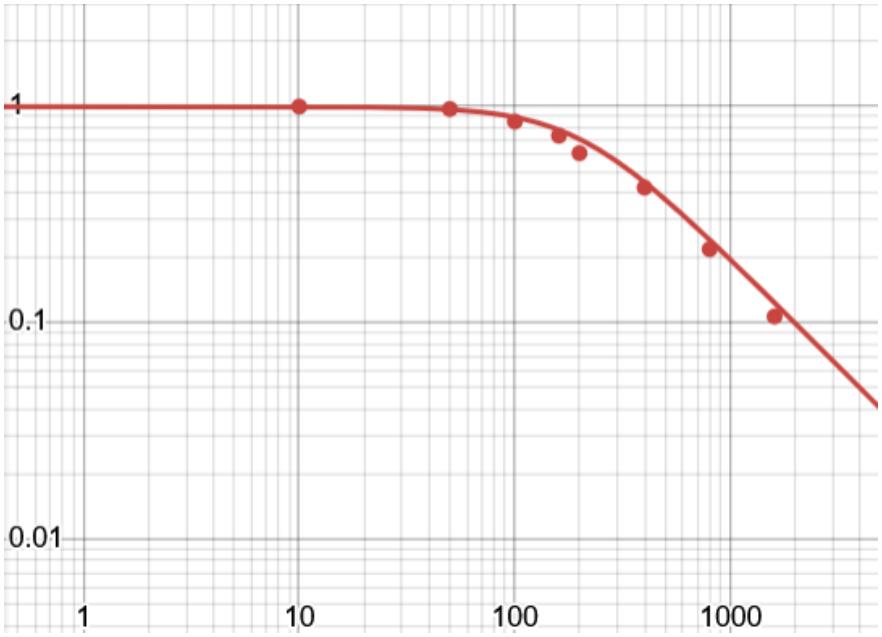


Figure 2: **Gain of the circuit**.
Bode plot where the x-axis is frequency, and the y-axis is the gain magnitude.

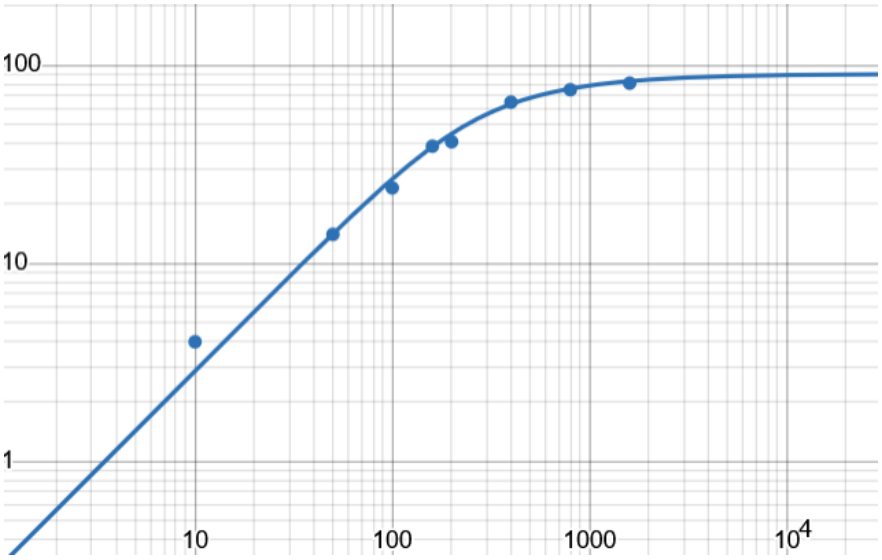


Figure 3: **Phase shift of the circuit**.
Bode plot where the x-axis is frequency, and the y-axis is the negative of the phase angle.

Notice how the magnitude starts to drop off at around 0.7 gain. This is where the cut-off for passing frequencies is, and what is considered low frequencies for the filter. The cut off is defined as where the magnitude is equal to $\frac{1}{\sqrt{2}}$. By setting $|H(j\omega)|$ equal to this number, we can find that the cut-off is $\frac{1}{RC}$ or $200Hz$. Similarly, the phase begins to plateau at the same frequency.

1.3 Conclusion

The plots for both the phase shift and gain magnitude are very close to the predictions. It's safe to say the plots are accurate predictors of the behavior of the filter. Interestingly, since this is on a log-log scale, if you zoom out from the parts of the graph where our measurements are, the functions actually look similar to a piecewise function in linear scales. For the gain, it's a constant one gain before the cutoff, then drops by $-0.1/$ decade. Similarly, for the phase, it plateaus at $-90°$ after the cut-off, but before has a slope of $-45°/$ decade.

2 Active High Pass Filter

2.1 Introduction

A high-pass filter only allows frequencies higher than a cut-off to pass with full effectiveness. And "active" means it includes active components like an op-amp. We will do something similar to the last section here and model the outputs compared to our calculated data.

I think we should see similar results to our last section, and that our data should match the predicting functions we can derive for our circuit.

2.2 Experiment

Our circuit is much more complex than the earlier section. The op amp system is designed to be an inverting op amp, and due to the ratio of resistors, would have a gain of $10x$ without the capacitor. The capacitor is $100nF$, R is $1k\Omega$ and R_f is $10k\Omega$. The circuit is pictured in Fig 4.

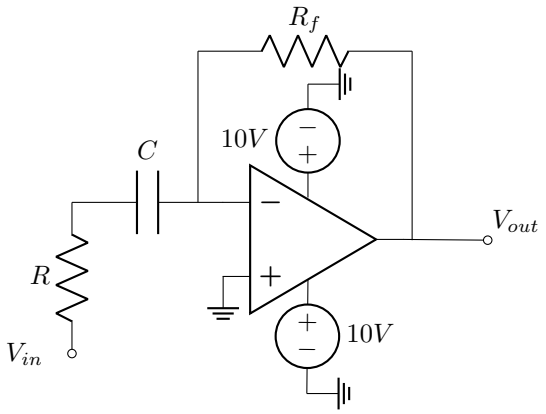


Figure 4: Active High Pass Filter

And we are collecting similar data to the previous section, just under a different set of frequencies. The data is in Table 2. The input voltage ended up being $1.09V$ peak-to-peak, which was too large to round off, so unlike the first section, the output amplitude will not equal the gain.

f	$V_{out}(V_{pp})$	$\lvert H(j\omega)\rvert$	$\lvert H(j\omega)\rvert$ (dB)	t_d (μs)	ϕ $(^\circ)$
$100Hz$	0.63	0.56	-5.04	2646	-95
$300Hz$	1.81	1.66	4.40	941.8	-102
$500Hz$	2.93	2.69	8.60	608.8	-110
$1kHz$	5.30	4.86	13.73	345.25	-124
$3kHz$	9.40	8.62	18.71	141.5	-153
$5kHz$	10.30	9.45	19.51	94.2	-170
$10kHz$	10.70	9.82	19.84	50.2	-181
$30kHz$	8.80	8.07	18.14	-12.7	137

Table 2: High Pass Data

I want to note that I don't believe the time delay and phase shift at $30kHz$ was correct, and I believe I made a measurement error. The opportunity to remake the measurement is no longer available to me, so I will ignore those points of data.

To create our predicting equation for the circuit, we can use KCL on the negative op amp node, which will allow us to directly relate the input and output voltages. After that, it only requires simple algebra to find our gain and simplify its expression.

$$\frac{V_{in}-0}{R+Z_C}=\frac{0-V_{out}}{R_f}, \text{ where, } Z_C=\frac{1}{j\omega C}$$
$$H(j\omega)=\frac{V_{out}}{V_{in}}=-\frac{R_f}{R+Z_C}=-\frac{R_f}{R+(\frac{1}{j\omega C})}$$
$$H(j\omega)=-\frac{R_f j\omega C}{Rj\omega C+1}$$

The function in the lab manual was expected to be something similar, its equivalent, but has $-R_f$ factored out.

$$H(j\omega)=-\frac{R_f}{R}\cdot\frac{Rj\omega C}{Rj\omega C+1}$$

The phase shift can be found similarly to the first section as well. Recall that $\theta=arctan(\frac{Imaginary}{Real})$, and that $\angle\frac{A}{B}=\angle A-\angle B$. As well as the angle of the products rule, $\angle A\cdot B=\angle A+\angle B$.

$$\phi^\circ=\angle-\frac{R_f}{R}+arctan(\frac{R\omega C}{0})-arctan(\frac{R\omega C}{1})$$
$$\phi^\circ=-180^\circ+90^\circ-arctan(R\omega C)$$
$$\phi^\circ=-90^\circ-arctan(R\omega C)$$

Our predicting data can now be graphed with these equations. Figures 5 and 6 depict them.

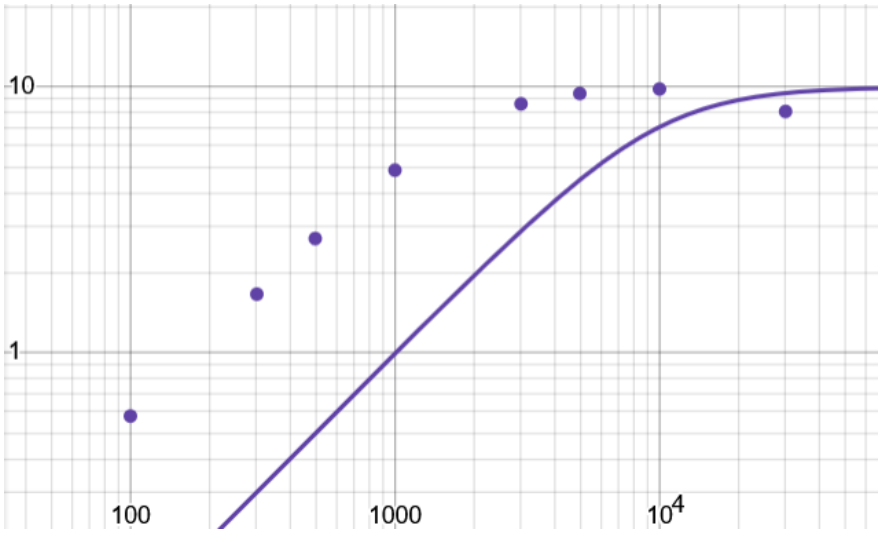


Figure 5: **Gain of the circuit**.
Bode plot where the x-axis is frequency, and the y-axis is the gain magnitude.

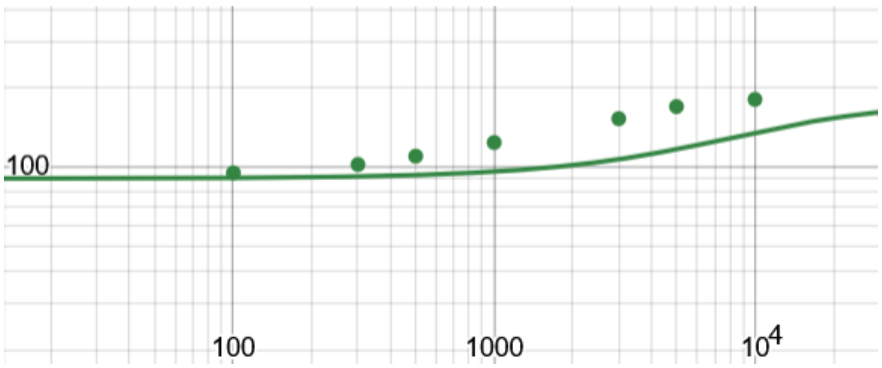


Figure 6: **Phase shift of the circuit**.
Bode plot where the x-axis is frequency, and the y-axis is the negative of the phase angle.

Strangely, the equations and the data appeared very off. After tweaking the data, I noticed that the actual measured data is better explained by a capacitance that is four times larger than what I believed was present in my capacitor. Alternatively, but less likely, both resistors are four times stronger than expected.

I have no very good explanation for how this happened since $300nF$ is a very large difference and should not have been possible with the components available in the lab. All the capacitors are multiples of 10 only. It could not have been a bad capacitor either, since it's expected that a capacitor drops capacitance instead of raising it when it goes bad.

I don't expect there to be that large of a capacitance in any of my test leads, either, since I have not observed that in any previous lab assignments and not even in this prior section. The Figures 7 and 8 depict these new predictions.

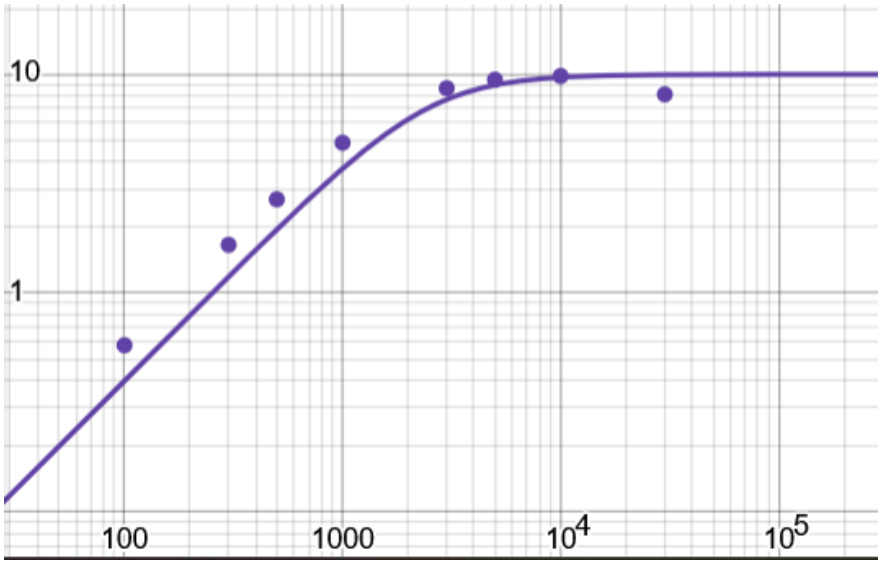


Figure 7: **Gain of the circuit with 400nF**.
Bode plot where the x-axis is frequency, and the y-axis is the gain magnitude.

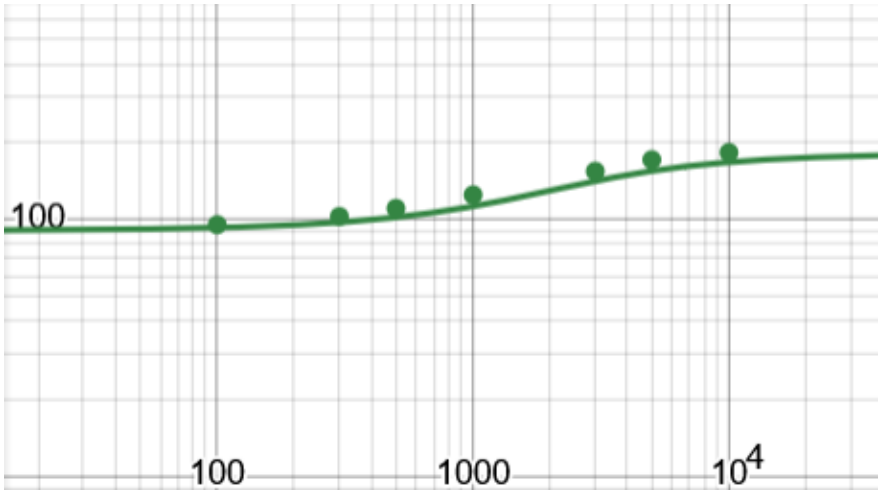


Figure 8: **Phase shift of the circuit with 400nF**.
Bode plot where the x-axis is frequency, and the y-axis is the negative of the phase angle.

2.3 Conclusion

This last section did not go smoothly, and I can only speculate as to how I managed to get such a large error. The original equations do not accurately match measured values; however, the ones where I assumed a higher capacitance fit very well. It may not seem like it, but for capacitors, the nano-scale is very important and sensitive; this was four times the expectation. In real-life applications, such a large error may be catastrophic since filters are used in a wide variety of devices. An easy place to find filters is anywhere with wireless communication, including radios, satellites, Bluetooth devices, cellphones, and telephone systems. If a filter were to fail working properly, the device would lose the capability to send and or receive signals. When batches of pre-made filters are tested, it is essential that the manufacturer tests them to avoid errors like this before they can make their way into essential systems.

One precaution I could have taken to avoid this would have been to use a capacitance meter to check if my capacitor was working properly. If it did not prevent the issue, it could also have narrowed the speculation on what caused the issue.